YUTONG GE

ROBOTICS SOFTWARE ENGINEER | ROBOT LEARNING · AUTONOMOUS SYSTEMS
Manchester, UK | github.com/GYT44 | 307172706@qq.com

PROFILE

MSc Robotics student at the University of Manchester with hands-on experience in ROS 2 control, humanoid robot simulation, reinforcement-learning workflows, embedded motion control and sensor-based localisation. Interested in robotics software and robot learning roles.

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB | Robotics: ROS 2, Isaac Lab, Isaac Sim, Unitree G1/H2, PID, SLAM testing
Tools & systems: Linux/Ubuntu, Raspberry Pi, MSPM0G3507, OpenMV, PTP, TDOA, Unity, SteamVR, Git

PROFESSIONAL EXPERIENCE

Robotics Algorithm Intern | Zhejiang SUPCON Information Co., Ltd. | Apr–Jun 2026

- Built and validated ROS 2 controller components for robot communication and motion control.
- Developed Unitree G1 locomotion tasks in Isaac Lab; tuned training configurations and validated behaviour in Isaac Sim.
- Supported Unitree H2 VR teleoperation experiments and SLAM navigation module testing.

SELECTED PROJECTS

Distributed Acoustic Source Localisation (TDOA) | Project Lead | Oct 2023–Nov 2025

- Led Raspberry Pi microphone-array development with PTP synchronisation and ALSA audio acquisition; implemented STFT, cross-correlation and TDOA localisation in MATLAB.
- Reported 0.2 ms synchronisation precision and 0.25 m 2D localisation accuracy; project generated patent and software-copyright outputs.

Autonomous Figure-Eight Navigation Robot | Embedded Software | Jul–Aug 2024

- Integrated MSPM0G3507, OpenMV, grayscale sensors and HWT101 gyroscope; implemented path recognition and dual-loop PID speed/steering control.
- Achieved figure-eight tracking, obstacle avoidance and fixed-point stopping; best recorded lap: 15.12 s.

Trimanual Tasks in Virtual Reality | Research Project | Jun–Aug 2025

- Developed six Unity/SteamVR three-handed tasks with PICO and VIVE Tracker; automated trajectory and task-performance data collection for MATLAB analysis.

Leo Rover Mobile Robotics | MSc Group Project | 2026–present

- Contributing to ROS 2 platform setup, onboard computing and hardware/sensor integration (ongoing).

EDUCATION

The University of Manchester | MSc Robotics | 2026–2027 (in progress)

Zhejiang Gongshang University | BEng Electronic Information Engineering | 2022–2026

International educational programme associated with the University of Sussex; average 88.6/100, top 15%.

SELECTED ACHIEVEMENTS

Gold Award, Zhejiang Challenge Cup Entrepreneurship Competition (2026); Silver Award, Zhejiang Challenge Cup Academic Science & Technology Competition (2025).